Software Requirements Specification

for

CANTEEN MANAGEMENT SYSTEM

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The product being specified is the **Canteen Management System (CMS)**, Version 1.0. This SRS describes the complete web-based system, including the customer ordering portal and the admin management interface. It covers the entire scope of the software required to digitize canteen operations, from menu browsing to QR-based order fulfillment.

1.2 Document Conventions

- **Typography:** This document is written in **Times New Roman**.
- **Priority Inheritance:** Priorities for higher-level requirements (High, Medium, or Low) are assumed to be inherited by all detailed requirements nested within them.
- **Requirement Granularity:** Every individual requirement statement is assigned its own unique sequence number or tag to ensure it is distinct and verifiable.
- **Highlighting: Bold text** is used for terms with special significance, such as user interface elements or key system actions.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

- **Developers:** Should read the entire document to understand technical constraints and system features.
- **Testers:** Should focus on Section 3 (System Features) and Section 5 (Nonfunctional Requirements) to develop test plans.
- **Canteen Admin (User):** Should focus on Section 2 (Overall Description) to understand the workflow and Section 4 (Interfaces) for the QR scanning process.
- **Sequence:** Readers should start with the Overview in Section 2 before proceeding to the specific technical details in Sections 3 and 4

1.4 Project Scope

The Canteen Management System is a web-based solution designed to replace the existing manual "counter-payment" system which currently results in disorganized queues.

- **Software Description:** The system provides a digital menu for customers and a management dashboard for the canteen representative.
- **Goals and Objectives:** The primary goal is to streamline the ordering process by allowing customers to book and pay online, receiving a QR code as a receipt.
- **Benefits:** This reduces the rush hour queue at the counters by enabling customers to pre-book the menu online.

1.5 References

Existing Canteen Manual Logbooks

2. Overall Description

2.1 Product Perspective

The Canteen Management System (CMS) is a new, self-contained web-based product designed to replace the existing manual ticketing system. While it is a standalone application, it functions as a digital layer over the physical canteen operations, transforming the workflow from a manual "pay-at-counter" model to a digital "order-and-verify" model. The system acts as a bridge between the customer and the canteen staff, managing the flow of order data, payments, and stock levels in real-time

2.2 Product Features

The major features of the CMS include:

- **Menu Browsing:** Real-time display of available food items and prices for customers.
- **Online Booking & Payment:** Interface for customers to select items, place orders, and complete digital payments.
- **QR Code Generation:** Automated creation of a digital receipt in the form of a QR code upon successful payment.
- **Admin Dashboard:** Tools for the canteen representative to add/remove menu items and update stock availability.
- **Order Verification:** A QR scanning feature for the admin to validate customer receipts and mark orders as fulfilled.

2.3 User Classes and Characteristics

- **Customer:** Generally who use the system to view the menu and book meals. They are expected to have basic technical expertise in operating a web browser on a smartphone or computer. This is the favoured user class for the ordering interface.

- **Admin (Canteen Representative):** A staff of canteen operations. This user requires higher privilege levels to modify the menu and stock. They must be able to operate the QR scanning interface to verify orders at the delivery point

2.4 Operating Environment

The software will operate in the following environment:

- **Platform:** Web-based (accessible via mobile and desktop browsers).
- **Client Side:** Compatible with modern web browsers (Chrome, Safari, Firefox).
- **Server Side:** Hosted on a web server with a connected database to store menu and transaction data.
- **Hardware:** Admin requires a device with a functional camera for QR code scanning

2.5 Design and Implementation Constraints

- **Technology:** The system must be developed as a responsive web application to ensure usability on mobile devices during peak canteen hours.

- **Security:** Payment processing must follow secure protocols to protect user financial data.

- **Connectivity:** Both the customer and admin require an active internet connection for real-time synchronization of stock and order status.

2.6 User Documentation

The following documentation will be delivered with the software:

- **Admin User Manual:** A guide covering menu management and the QR verification process.
- **Online Help:** A brief "How to Order" guide integrated into the customer web interface.

2.7 Assumptions and Dependencies

- **Assumptions:** It is assumed that all customers have access to a smartphone with internet connectivity while at the canteen. We also assume the digital payment gateway remains operational during canteen hours.

- **Dependencies:** The system depends on a reliable third-party payment API for processing transactions.

3. System Features

3.1 Menu Management (Admin)

3.1.1 Description and Priority

This feature allows the canteen representative to maintain the digital menu.

Priority: High

3.1.2 Stimulus/Response Sequences

1. **Stimulus:** Admin adds a new dish with price and category.
2. **Response:** System saves the item and displays it on the customer-facing menu.

3.1.3 Functional Requirements

- **REQ-1:** The system shall allow the Admin to upload images for menu items.
- **REQ-2:** The system shall allow the Admin to update prices in real-time.
- **REQ-3:** The system shall allow the Admin to toggle "Sold Out" status for any item.

3.2 Order Booking & Payment (Customer)

3.2.1 Description and Priority

This allows customers to select items and pay digitally.

Priority: High

3.2.2 Stimulus/Response Sequences

Stimulus: Customer adds items to a virtual cart and proceeds to pay.
Response: System redirects to a payment gateway and confirms the transaction.

3.2.3 Functional Requirements

REQ-4: The system shall calculate the total bill including any applicable taxes.
REQ-5: The system shall handle payment failures by notifying the user and not deducting stock.

3.3 QR Receipt & Verification

3.3.1 Description and Priority

The core mechanism for order fulfillment and reducing queue mess.

Priority: High

3.3.2 Stimulus/Response Sequences

Stimulus: System generates a QR code upon successful payment.
Response: Customer presents QR to Admin; Admin scans and verifies.

3.3.3 Functional Requirements

REQ-6: The system shall generate a unique, non-reusable QR code for every order.
REQ-7: The QR code must contain the Order ID and a summary of items purchased

3.4 Inventory and Stock Tracking

3.4.1 Description and Priority

This feature manages the availability of food items based on real-time stock levels.

Priority: Medium.

3.4.2 Stimulus/Response Sequences

Stimulus: Successful payment confirmation from a customer.

Response: System decrements the item quantity in the database.

Stimulus: Stock count for an item reaches zero.

Response: System updates the item status to "Sold Out" on the customer menu.

3.4.3 Functional Requirements

REQ-8: The system shall automatically reduce the available stock count for an item upon a successful transaction.

REQ-9: The system shall disable the "Book" or "Add to Cart" option for any item whose stock count is zero.

REQ-10: The system shall allow the Admin to manually update stock quantities through the management dashboard.

- **REQ-11:** The system shall provide an error message to the customer if an item in their cart becomes unavailable before payment is completed

4. External Interface Requirements

4.1 User Interfaces

- **Customer Interface:** A responsive web portal featuring a categorized menu, a virtual shopping cart, and a secure checkout page.
- **Admin Interface:** A password-protected dashboard for menu management and a dedicated "Verification Mode".
- **Standards:** The system shall follow a consistent color scheme and layout, with a standard navigation bar and a "Help" button available on every screen.
- **Layout Constraints:** The interface must be optimized for mobile viewing, ensuring buttons are touch-friendly for users in a physical canteen environment.
- **Error Messages:** All validation errors (e.g., "Invalid QR" or "Payment Failed") must be displayed in a clear, standard red notification box at the top of the screen

4.2 Hardware Interfaces

- **Scanning Device:** The system shall interface with the camera of the Admin's device (smartphone, tablet, or laptop) to capture and decode QR code images.
- **Customer Device:** The system assumes a standard smartphone or PC with a screen capable of displaying a high-contrast QR code for scanning.
- **Communication:** Interaction between the software and the camera hardware will be handled via standard web browser APIs (such as the MediaDevices API).

4.3 Software Interfaces

- **Database:** The system shall connect to a SQL or NoSQL database to store and retrieve menu items, user credentials, and transaction history.
- **Payment Gateway:** The system shall interface with a third-party payment API (like Stripe, Razorpay, or PayPal) to process secure financial transactions.
- **Operating Systems:** The web application must be compatible with Android, iOS, Windows, and macOS through modern web browsers

4.4 Communications Interfaces

- **Protocol:** All data transfer between the client and server shall use the HTTPS protocol to ensure encryption and security.
- **QR Generation:** The system will use a communication standard to generate and display the QR code as a browser-rendered image.
- **Notifications:** The system shall use SMTP for sending optional email receipts or web sockets for real-time order status updates to the Admin dashboard.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** The system shall load the menu within 2 seconds under normal network conditions to prevent user frustration.
- **Concurrent Users:** The web application must support at least 50 concurrent customer sessions during peak canteen hours without a degradation in performance.
- **Real-time Updates:** Stock level changes made by the Admin must reflect on all active customer browsers within 5 seconds.

5.2 Safety Requirements

- **Data Loss:** The system shall implement automated database backups every 24 hours to prevent loss of transaction history.
- **Order Integrity:** The system must ensure that a QR code is invalidated immediately after a single successful scan to prevent "double-claiming" of food items.
- **Transaction Safety:** In the event of a system crash during payment, the system must ensure no funds are deducted without a corresponding order being recorded

5.3 Security Requirements

- **Authentication:** The Admin dashboard must be protected by a secure login requiring a username and a strong password.
- **Privacy:** Customer payment details (like card numbers) shall not be stored in the CMS database; they must be handled exclusively by the third-party payment provider.
- **Authorization:** Only users with Admin privileges shall have access to menu modification and stock update functions.
- **Encryption:** All communication between the client and the server must be encrypted using SSL/TLS (HTTPS) to protect data in transit.

5.4 Software Quality Attributes

- **Usability:** The customer interface shall be designed for "one-hand" operation, as users may be standing or walking in the canteen.
- **Availability:** The system should be available 99% of the time during canteen operating hours.
- **Reliability:** The QR generation and scanning logic must have a success rate of 99.9% to avoid manual intervention at the counter.
- **Adaptability:** The web application must be responsive, automatically adjusting its layout for smartphones, tablets, and desktop screens.

6. Other Requirements

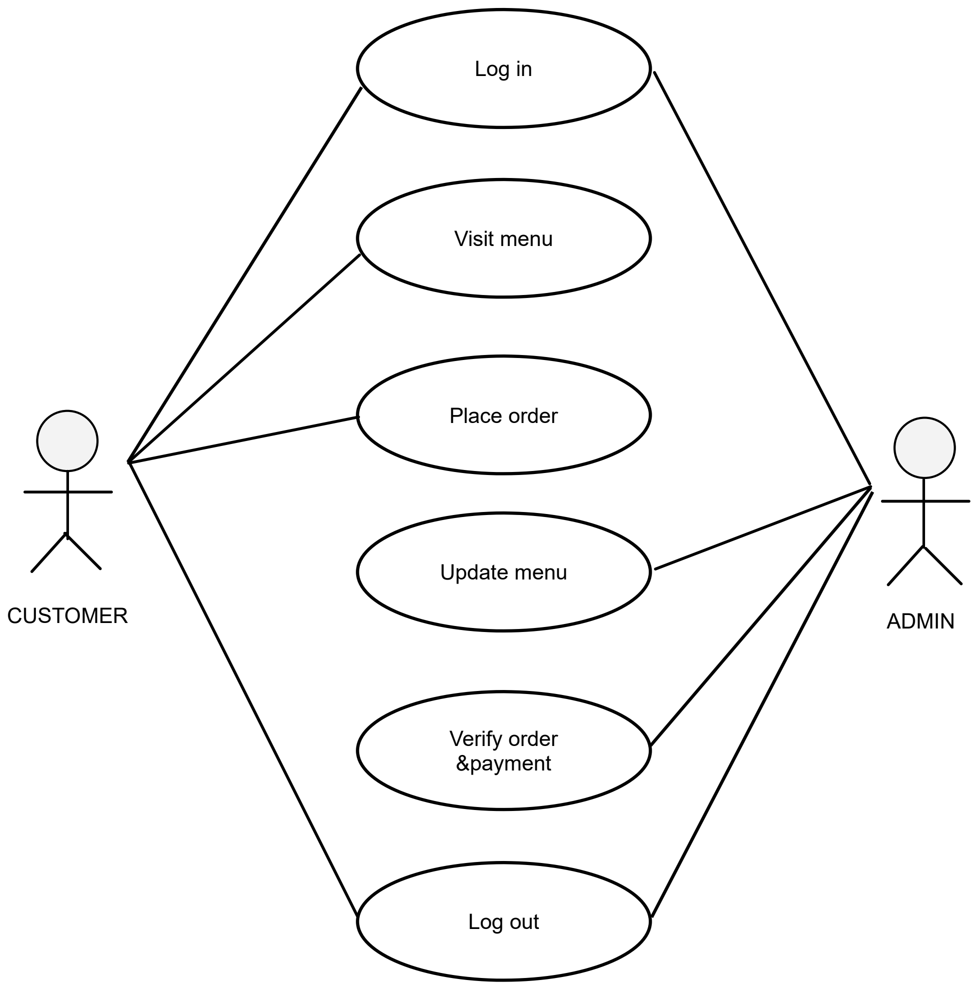
Database Requirements: The system shall maintain a permanent log of all transactions for at least one academic year for audit purposes

Appendix A: Glossary

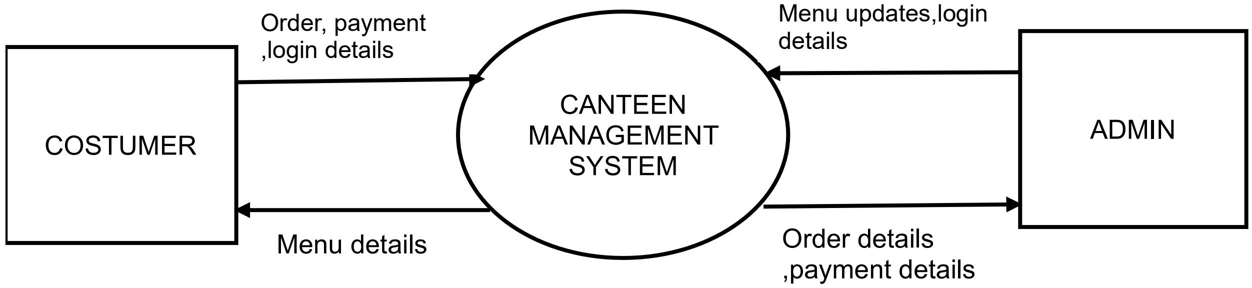
Admin: The canteen representative responsible for menu updates and order verification.
Customer: The student or staff member using the system to purchase food.

Appendix B: Analysis Models

Use case diagram



Data flow Diagram



Appendix C: Issues List

Payment Gateway Choice: Which specific third-party provider will be used (e.g., Razorpay, Stripe)?.
Offline Mode: How should the system handle a sudden internet outage at the canteen counter?.
Refund Policy: What is the automated process if a customer cancels an order after payment?.
Hardware Specs: Confirming if the Admin device camera meets the resolution requirements for scanning in low-light canteen conditions